SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

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## 1. Experiment Title:

Use string functions and regex for email validation, data extraction, and text analysis.

## 2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

## 3. Experiment Program Code:

*Experiment 6.html*

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>String Methods & Regular Expressions</title>
    <link rel="stylesheet" href="Experiment 6.css">
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
</head>
<body>
    <div class="container">
        <h1>String Methods & Regular Expressions</h1>
        <div class="box">
            <div class="input-group">
                <label for="paragraph"><b>Enter a paragraph:</b></label>
                <textarea id="paragraph" rows="5">Javascript is a powerful programming language. It
is widely used for web development.</textarea>
            </div>

            <div class="input-group">
                <label for="email"><b>Enter Email:</b></label>
                <input type="text" id="email" placeholder="Enter your email">
            </div>

            <button onclick="processString()">Process</button>
            <div id="output"></div>
        </div>

        <footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; border-
top: 1px solid lightgray; padding-top: 15px;">
            Made by Rohan Laharwani PRN - 24070521055
        </footer>
    </div>

    <script src="Experiment 6.js"></script>
</body>
</html>
```

*Experiment 6.css*

```
:root {
    --primary: #2563eb;
    --primary-hover: #1d4ed8;
    --bg-main: #f8fafc;
    --surface: #ffffff;
    --text-primary: #1e293b;
    --text-secondary: #64748b;
    --border-color: #e2e8f0;
    --radius: 10px;
    --shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.05), 0 2px 4px -2px rgba(0, 0, 0, 0.05);
}

* {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}

body {
    background-color: var(--bg-main);
    color: var(--text-primary);
    font-family: 'Inter', system-ui, -apple-system, sans-serif;
    line-height: 1.5;
    padding: 40px 20px;
    display: flex;
    justify-content: center;
}

.container {
    width: 100%;
    max-width: 800px;
    background-color: var(--surface);
    padding: 32px;
    border: 1px solid var(--border-color);
    border-radius: 16px;
    box-shadow: var(--shadow);
}

.app-header {
    text-align: center;
    margin-bottom: 28px;
}

.app-header  h1  {
    color: #1e3a8a;
    font-size: 26px;
    font-weight: 700;
    margin-bottom: 6px;
}

.subtitle {
    color: var(--text-secondary);
    font-size: 14px;
}

.box {
    display: flex;
    flex-direction: column;
    gap: 18px;
}

.input-group {
    display: flex;
    flex-direction: column;
    gap: 6px;
}

label {
```

```
    font-size: 14px;
    color: var(--text-primary);
}

input, textarea {
    padding: 12px;
    width: 100%;
    border: 1px solid var(--border-color);
    border-radius: var(--radius);
    font-family: inherit;
    font-size: 14px;
    outline: none;
    transition: all 0.2s ease;
    background-color: #f8fafc;
}

input:focus, textarea:focus {
    border-color: var(--primary);
    background-color: #ffffff;
    box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.15);
}

.action-group {
    margin-top: 4px;
}

button {
    padding: 12px 24px;
    background-color: var(--primary);
    color: white;
    border: none;
    border-radius: var(--radius);
    font-size: 15px;
    font-weight: 600;
    cursor: pointer;
    transition: background-color 0.2s ease, transform 0.1s ease;
}

button:hover {
    background-color:  var(--primary-hover);
}

button:active {
    transform: scale(0.98);
}

.output-container {
    margin-top: 10px;
}

.results-grid {
    display: flex;
    flex-direction: column;
    gap: 16px;
}

.card {
    background: #f8fafc;
    border: 1px solid var(--border-color);
    border-left: 4px solid var(--primary);
    border-radius: var(--radius);
    padding: 16px;
    transition: transform 0.15s ease, box-shadow 0.15s ease;
}

.card:hover {
    transform: translateY(-2px);
    box-shadow: var(--shadow);
}
```

```
.card h3 {
    color: #1e3a8a;
    font-size: 15px;
    font-weight: 600;
    margin-bottom: 8px;
}

.card p {
    font-size: 14px;
    color: var(--text-primary);
    word-break: break-word;
}

.card small {
    display: block;
    margin-top: 6px;
    color: var(--text-secondary);
    font-size: 12px;
}

.code-block {
    font-family: monospace;
    background: #e2e8f0;
    padding: 6px 10px;
    border-radius: 6px;
    font-size:  13px;
    color: #0f172a;
}

.reversed-text {
    font-family: monospace;
    color: #475569;
    letter-spacing: 1px;
}

.badge {
    display: inline-block;
    padding: 4px 10px;
    border-radius: 20px;
    font-size: 12px;
    font-weight: 600;
}

.badge.success {
    background-color: #dcfce7;
    color: #166534;
}

.badge.danger {
    background-color: #fee2e2;
    color: #991b1b;
}

.error-card {
    border-left-color: #ef4444;
    background-color: #fef2f2;
}

.error-msg {
    color: #b91c1c;
    font-weight: 500;
}

.app-footer {
    text-align: center;
    margin-top: 32px;
    font-size: 13px;
    color: var(--text-secondary);
```

```
    border-top: 1px solid var(--border-color);
    padding-top: 16px;
}
```

*Experiment 6.js*

```
function processString() {
    let paragraph = document.getElementById("paragraph").value;
    let email = document.getElementById("email").value;

    let words = paragraph.split(/\s+/);

    let vowels = paragraph.match(/[aeiou]/gi);
    let vowelCount = vowels ? vowels.length : 0;

    let replacedParagraph = paragraph.replace(
        /JavaScript/gi,
        "JavaScript Programming"
    );

    let searchWord = "powerful";
    let position = paragraph.indexOf(searchWord);

    let emailRegex =
    /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;

    let emailResult;

    if(emailRegex.test(email)) {
        emailResult = "Valid Email Address";
    } else {
        emailResult = "Invalid Email Address";
    }
    let emailText =
    "For queries contact student@example.com or admin@college.edu";

    let extractedEmails = emailText.match(
        /[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}/g
    );

    let reeversedParagraph = paragraph.split("").reverse().join("");

    document.getElementById("output").innerHTML = `
    <h3>1. Original Paragraph</h3>
    <p>${paragraph}</p>

    <h3> 2. split() - Words </h3>
    <p> ${words.join(",")}</p>

    <h3> 3. match() - Vowels </h3>
    <p> ${vowels ? vowels.join(", ") : "NO VOWELS FOUND!"}</p>

    <h3> 4. Vowel Count </h3>
    <p> Total number of vowels: <b>${vowelCount}</b></p>

    <h3> 5. replace() - Replace Text </h3>
    <p> ${replacedParagraph} </p>

    <h3> 6. indexOf() - Search Word </h3>
    <p>
        Position of "<b>${searchWord}</b>":
        <b>${position}</b>
    </p>

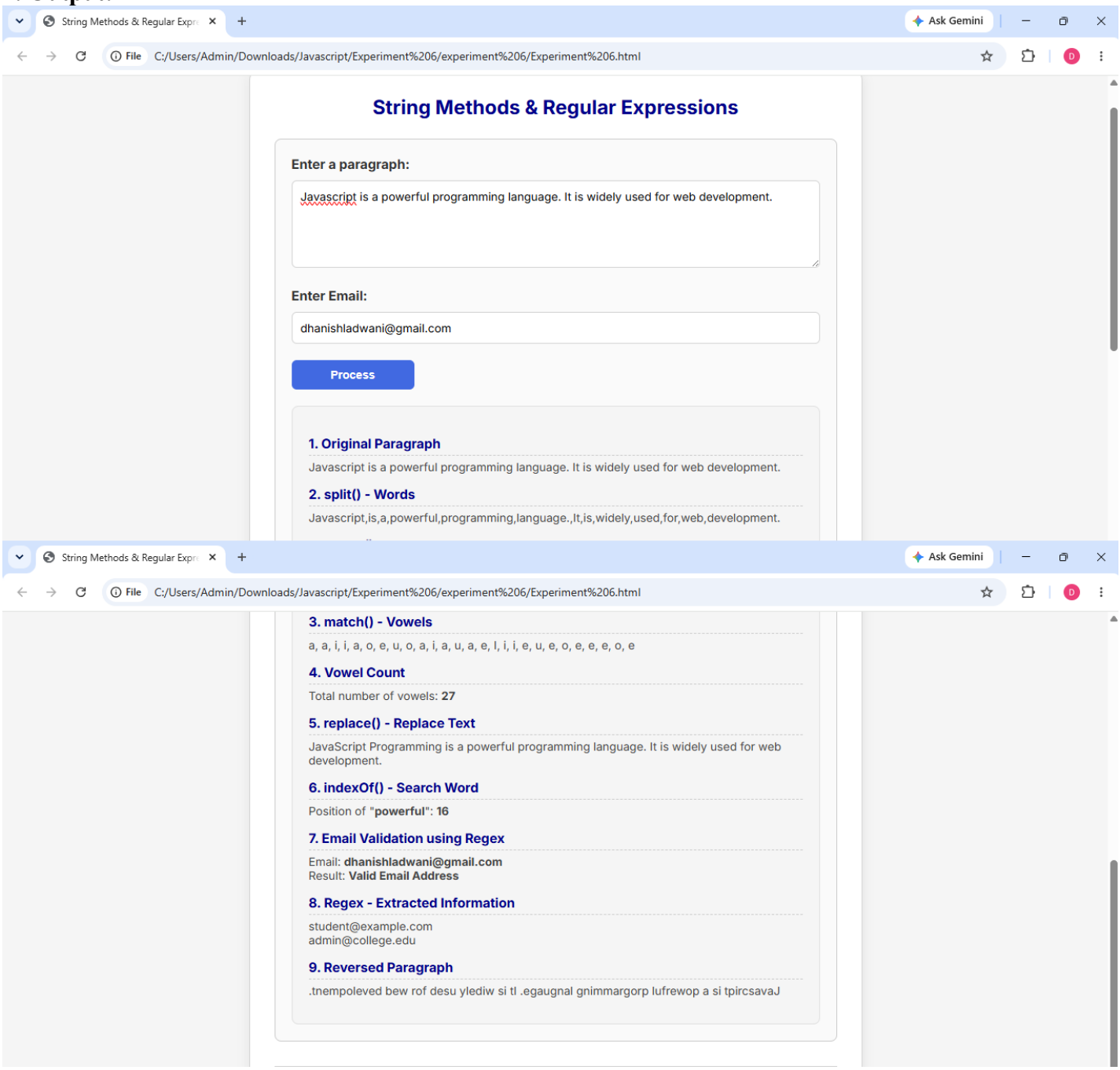
    <h3> 7. Email Validation using Regex </h3>
    <p>
```

```
        Email: <b>${email}</b><br>
        Result: <b>${emailResult}</b>
    </p>

    <h3> 8. Regex - Extracted Information </h3>
    <p>
        ${extractedEmails ? extractedEmails.join("<br>") : "NO EMAIL ADDRESS FOUND!"}
    </p>

    <h3> 9. Reversed Paragraph </h3>
    <p> ${reeversedParagraph} </p>
    `;
}
```

## 4. Output:



## 5. Result/Conclusion:

A web application was built to demonstrate JavaScript string methods and regular expressions. The split() method was used to break a paragraph into words, match() was used with a regular expression to extract vowels and count them, and replace() was used to substitute a keyword throughout the text. The indexOf() method located the position of a specific word, while regular expressions validated an entered email address and extracted all valid email addresses from a block of text. The split-reverse-join technique was also used to reverse the paragraph. This experiment reinforced string manipulation and pattern matching using regular expressions in JavaScript.